

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1: Experiments

**COURSE NAME:** Cloud Computing and  
Big Data Analytics

**COURSE CODE:** CSA1504

---

### COURSE OUTCOME COVERED

**CO3:** *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)  
Question Count: 4

**Total Marks: 40**

Duration : 60 Minutes

---

### Code of Conduct

I S Jefna Princy – 192572058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: \_\_\_\_\_

### Experiments Rubric

| Criteria                 | Weightage | Level 4 - Exemplary                                   | Level 3 - Proficient              | Level 2 - Developing   | Level 1 - Beginning             |
|--------------------------|-----------|-------------------------------------------------------|-----------------------------------|------------------------|---------------------------------|
| Experimental Design      | 25%       | Design is systematic and technically strong           | Mostly correct design             | Basic design with gaps | Poorly designed activity        |
| Use of Tools / Platforms | 20%       | Appropriate and effective use of cloud tools          | Good tool usage with minor issues | Limited tool usage     | Incorrect or missing tool usage |
| Application of Concepts  | 20%       | Concepts are applied accurately to practical tasks    | Mostly accurate application       | Partial application    | Weak application                |
| Analysis and Output      | 20%       | Strong analysis and meaningful outcome interpretation | Good analysis with minor gaps     | Basic analysis         | Little or no analysis           |
| Documentation            | 15%       | Clear, structured, and complete documentation         | Mostly complete documentation     | Partially complete     | Poor documentation              |

Total Marks Awarded:

## Assessment Tasks-2

### Experiments

**Experiment 1:** Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

**Experiment 2:** Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

**Experiment 3:** Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

**Experiment 4:** Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

## Assessment Tasks-2

### Experiment 1

Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

#### Aim

To create a cloud-based **Library Book Reservation System for SIMATS Library** using **Zoho Creator** and demonstrate the **Software as a Service (SaaS)** model.

#### Procedure

1. Log in to **Zoho Creator** using a valid user account.
2. Click **Create New Application** and name it **Hospital Management System**.
3. Create a **Patient Details** form and add fields such as Patient ID, Patient Name, Age, Gender, Phone Number, Address, Blood Group, and Disease.
4. Create a **Doctor Details** form and add fields such as Doctor ID, Doctor Name, Specialization, Phone Number, Email, and Consultation Fee.
5. Create an **Appointment Details** form and add fields such as Appointment ID, Patient (Lookup), Doctor (Lookup), Appointment Date, Appointment Time, and Appointment Status.
6. Configure the **Lookup** fields to link the Appointment Details form with the Patient Details and Doctor Details forms.
7. Save all the forms and allow Zoho Creator to automatically generate the corresponding reports.
8. Enter sample patient, doctor, and appointment records into the application.
9. Verify that all the entered records are displayed correctly in their respective reports.
10. Access the application through a web browser to demonstrate **Software as a Service (SaaS)** using the cloud platform.

### Patient Details:

Doctor's details:

Appointment:

**Result:** Thus, a cloud-based **Library Book Reservation System for SIMATS Library** was successfully created using **Zoho Creator**. The application stores book details, student information, and reservation records in the cloud and can be accessed through a web browser, thereby demonstrating the **Software as a Service (SaaS)** model.

## Experiment 2

Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

### Aim

To create a cloud-based **Online Super Market** application using **Zoho Creator** and demonstrate the **Software as a Service (SaaS)** model.

### Procedure

1. Log in to **Zoho Creator** using a valid user account.
2. Create a new application and name it **Online Super Market**.
3. Create a **Customer Details** form and add the required fields to store customer information.
4. Create an **Item Classification** form and add the required fields to classify supermarket items.
5. Create an **Item Details** form and add the required fields to maintain item information.
6. Create an **Order Details** form to record customer orders.
7. Configure **Lookup** fields to link the item details with item classification and the order details with customer and item details.
8. Save all the forms and allow Zoho Creator to generate the corresponding reports.
9. Enter sample customer, item, and order records into the application.
10. Verify that all records are stored correctly and displayed in the reports.
11. Access the application through a web browser to demonstrate that it is available as a cloud-based **Software as a Service (SaaS)**.

## Output

### Customer Details:

Online Super Market

Customer Details

Customer Details

Customer Details Report

Item Classification

Item Details

Order Details

Customer Details Report

Q

+

...

| Customer Name | Phone           | Address                                                               | Email             | Customer ID | Gender |
|---------------|-----------------|-----------------------------------------------------------------------|-------------------|-------------|--------|
| Rohith        | +91897326927382 | 23, Banjara Hills Road No. 12, Ernakulam, Karnataka, 3874632, Angola  | rohith@gmail.com  | 4           | Male   |
| judith        | +917834239283   | 78, Anna Salai, T. Nagar, Coimbatore, Tamil nadu, 3743764, Algeria    | judith@gmail.com  | 3           | Male   |
| kenisha       | +919324839272   | 45, MG Road, Ashok Nagar, Kanyakumari, Kerala, 682030, India          | kenisha@gmail.com | 2           | Female |
| Jefna         | +918347389229   | 12, Green Park Road, Kakkanad, Coimbatore, Tamil nadu, 6743223, India | jefna@gmail.com   | 1           | Female |

### Item Classification:

Online Super Market

OSM Online Super Market

Customer Details

Item Classification

Item Classification

Item Classification Report

Item Details

Order Details

Trial expires in 10 days Upgrade

Edit this application

Help

Item Classification Report

Category ID

Category Name

Description

|   |                     |                                                                    |
|---|---------------------|--------------------------------------------------------------------|
| 5 | Bakery              | read, cakes, buns, cookies, and other bakery products.             |
| 4 | Snacks              | Chips, biscuits, chocolates, and ready-to-eat snack items.         |
| 3 | Grocery             | Rice, wheat, pulses, flour, sugar, and essential grocery products. |
| 2 | Dairy Products      | Milk, curd, butter, cheese, and other dairy items.                 |
| 1 | Fruits & Vegetables | Fresh fruits and vegetables available for daily use.               |

### Item Details:

Online Super Market

Online Super Market

Customer Details

Item Classification

Item Details

Item Details

Item Details Report

Order Details

Trial expires in 10 days Upgrade

Edit this application

Help

Item Details Report

Item ID

Item Name

Category

Brand

Price

Stock Quantity

Availability

5

Biscuits

Snacks

Parle-G

₹ 50.00

0

Out of Stock

4

Shampoo

Grocery

Clinic Plus

₹ 65.00

10

In Stock

3

apple(1kg)

Fruits & Vegetables

₹ 100.00

0

Out of Stock

2

Milk (1 L)

Dairy Products

Amul

₹ 35.00

20

In Stock

1

Potato Chips

Snacks

Lay's

₹ 20.00

10

In Stock

## Order Details:

Online Super Market

Trial expires in 10 days Upgrade Edit this application Help

Online Super Market

Order Details Report

Customer Details

Item Classification

Item Details

Order Details

Order Details Report

| Order ID | Customer Details | Item Details | Quantity | Order Date  | Total Amount | Payment Method   |
|----------|------------------|--------------|----------|-------------|--------------|------------------|
| 4        | 4                | 5            | 3        | 13-Aug-2026 | ₹ 220.00     | Credit Card      |
| 3        | 3                | 3            | 1        | 06-Aug-2026 | ₹ 100.00     | Cash on Delivery |
| 2        | 2                | 2            | 3        | 04-Aug-2026 | ₹ 100.00     | Cash on Delivery |
| 1        | 1                | 1            | 2        | 04-Aug-2026 | ₹ 40.00      | UPI              |

## Result

Thus, a cloud-based **Online Super Market** application was successfully created using **Zoho Creator**. The application stores customer, item classification, item, and order details in the cloud and can be accessed through a web browser, thereby demonstrating the **Software as a Service (SaaS)** model.

## Experiment 3

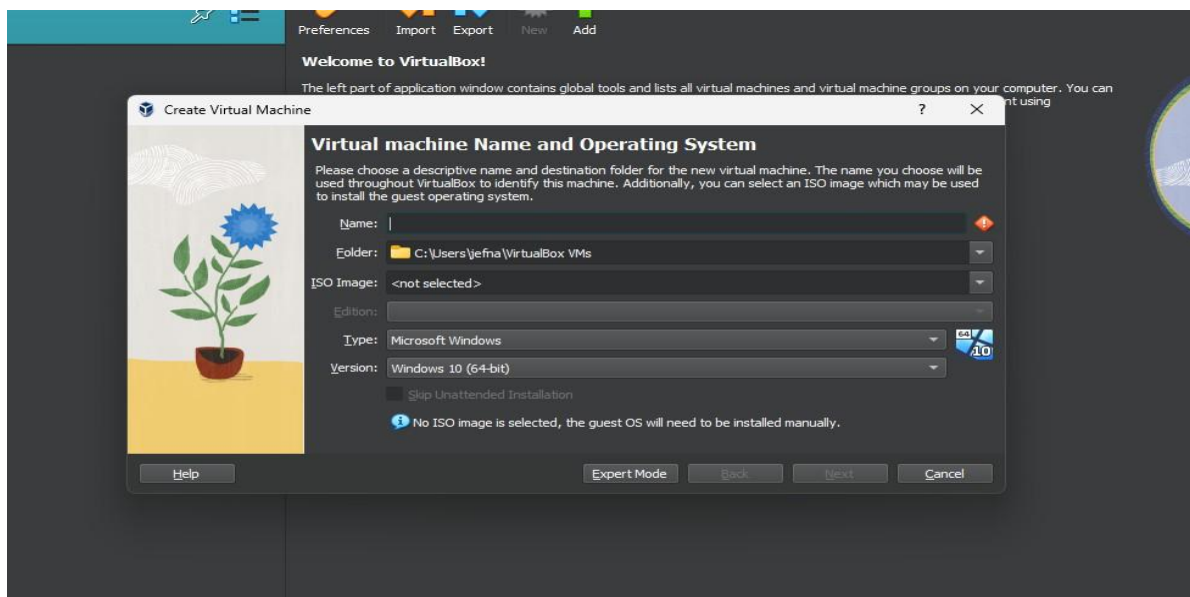
### Aim

To demonstrate virtualization by installing a Type-2 Hypervisor, creating and configuring a virtual machine with 3 CPU cores, 10 GB RAM, and 10 GB virtual storage using Oracle VirtualBox.

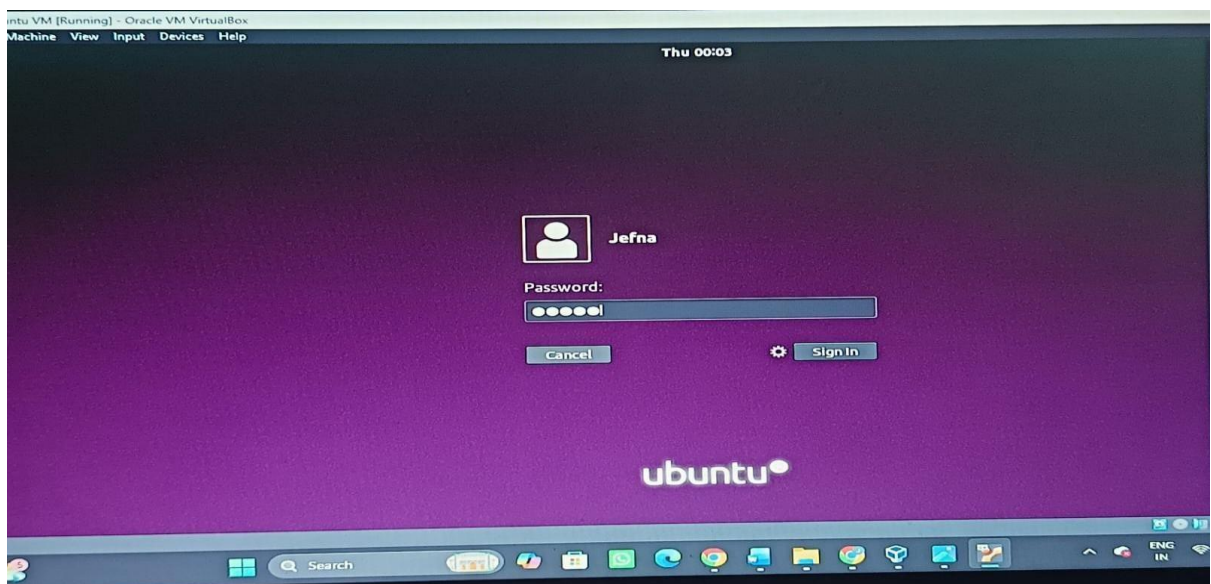
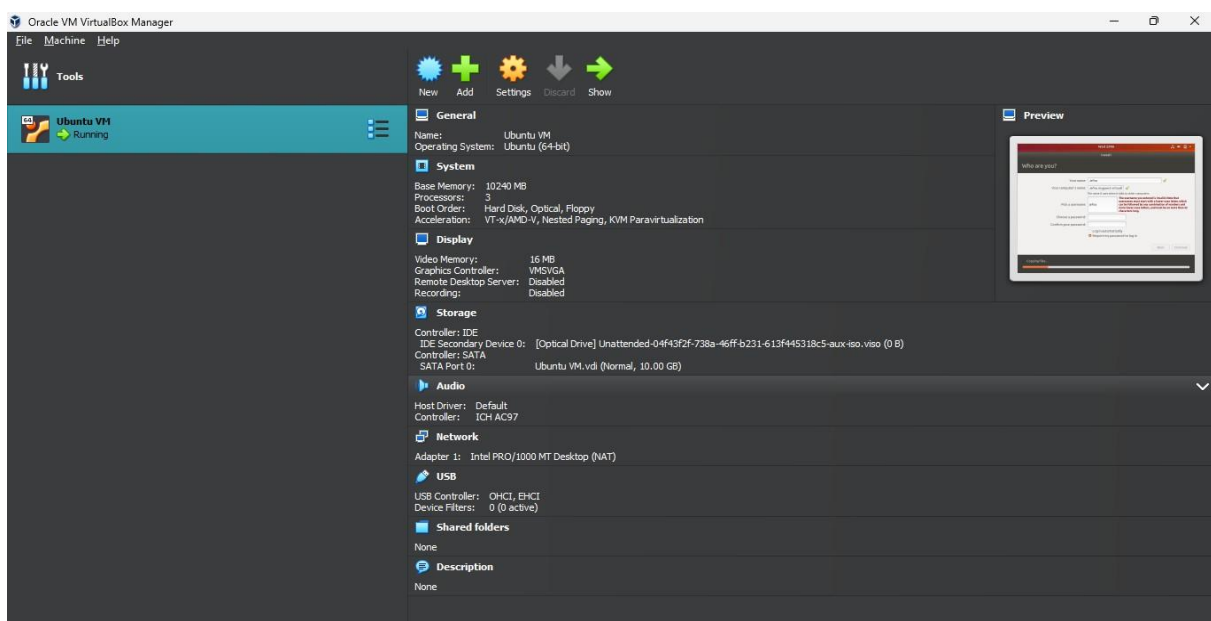
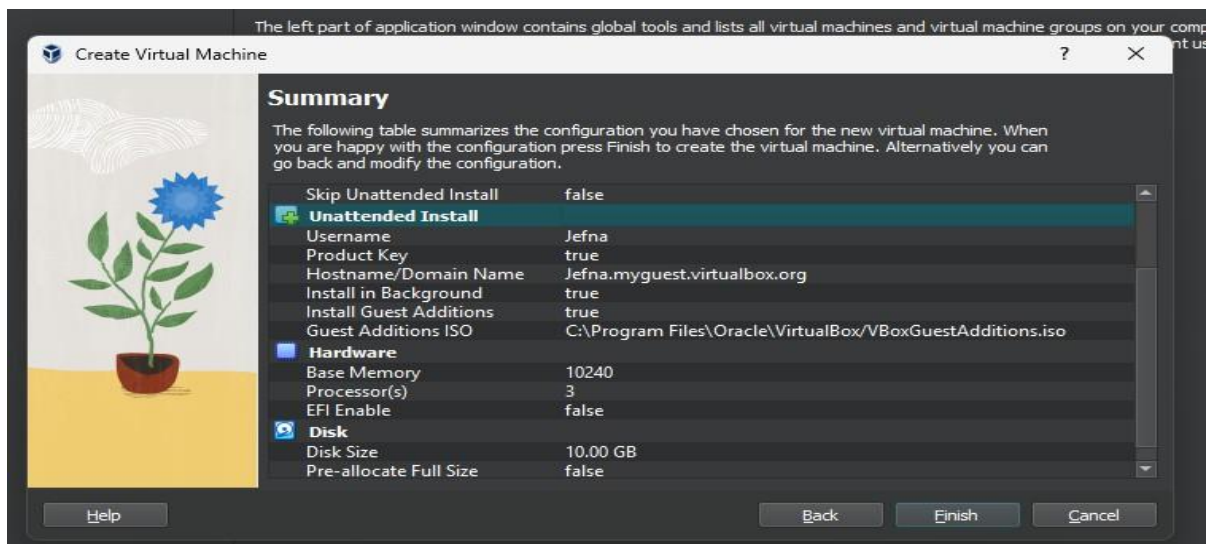
### Procedure

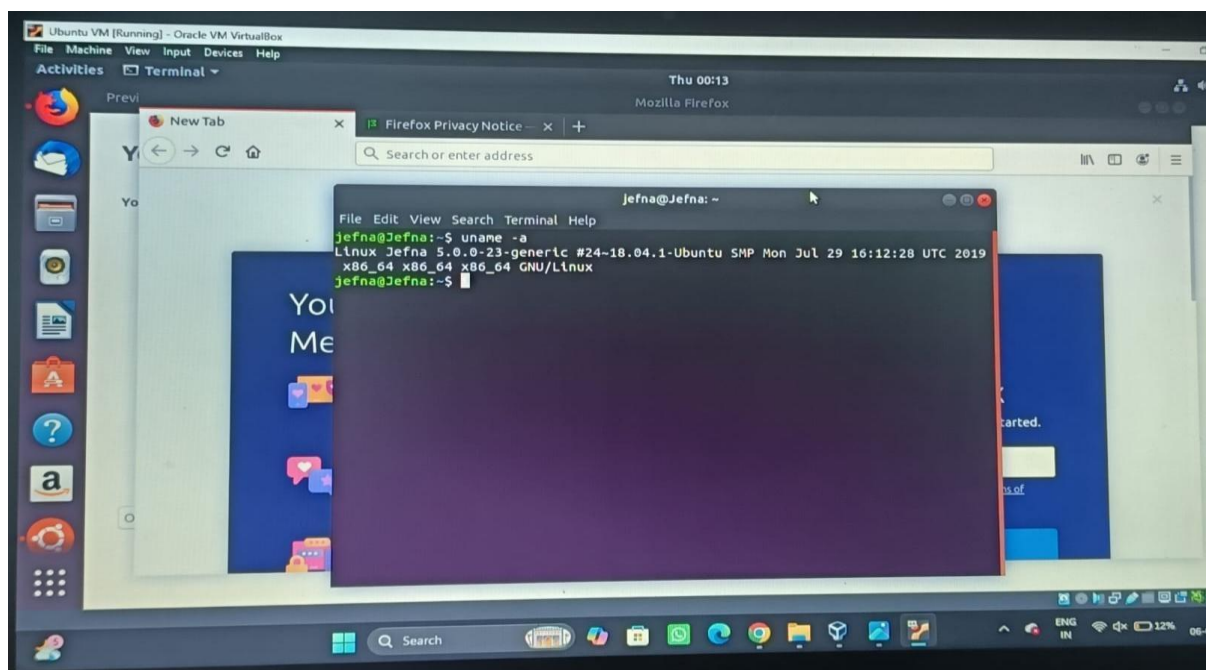
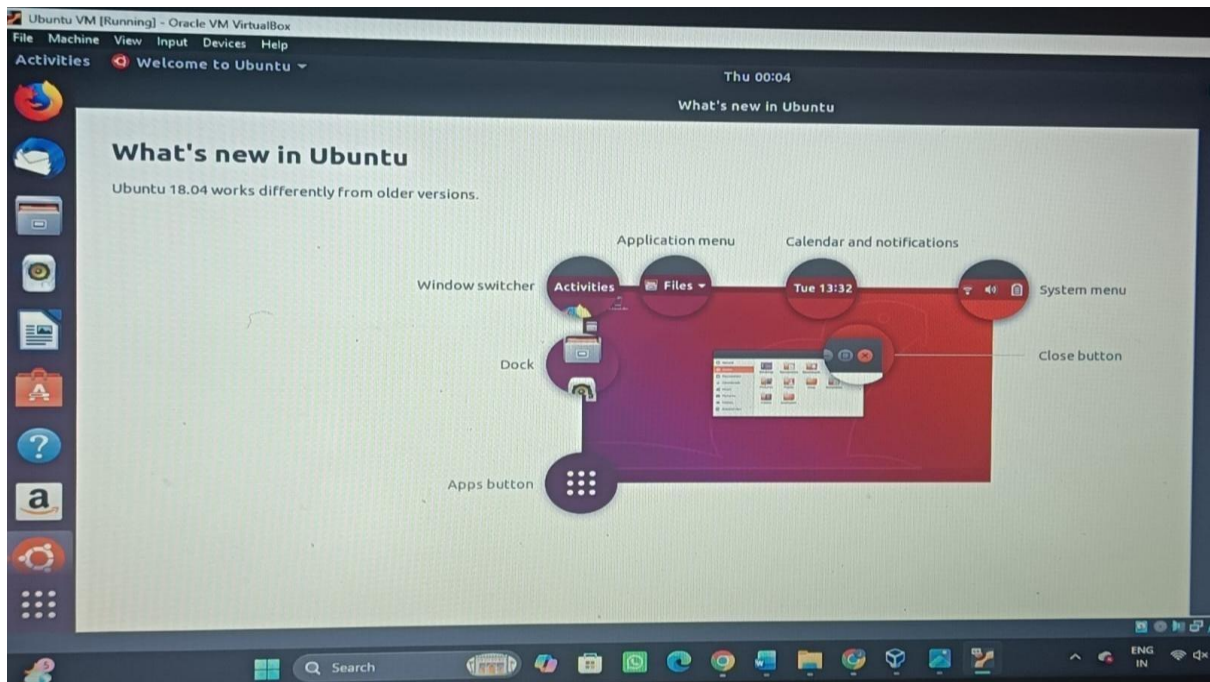
1. Download and install Oracle VirtualBox on the host operating system.
2. Download the Ubuntu Linux ISO image.
3. Open VirtualBox and create a new virtual machine.
4. Select Linux as the operating system.
5. Allocate 10 GB (10240 MB) RAM.
6. Assign 3 CPU cores to the virtual machine.
7. Create a 10 GB virtual hard disk.
8. Attach the Ubuntu ISO file to the virtual machine.
9. Start the virtual machine and install Ubuntu Linux.
10. Verify that the virtual machine boots successfully.

Output:









## Result

Thus, Type-2 virtualization was successfully demonstrated by installing Oracle VirtualBox on the host operating system. A virtual machine was created and configured with 3 CPU cores, 10 GB RAM, and a 10 GB virtual hard disk, and the Ubuntu Linux operating system was installed successfully. Hence, the virtual machine was created and executed successfully.

## **Experiment 4**

**Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM**

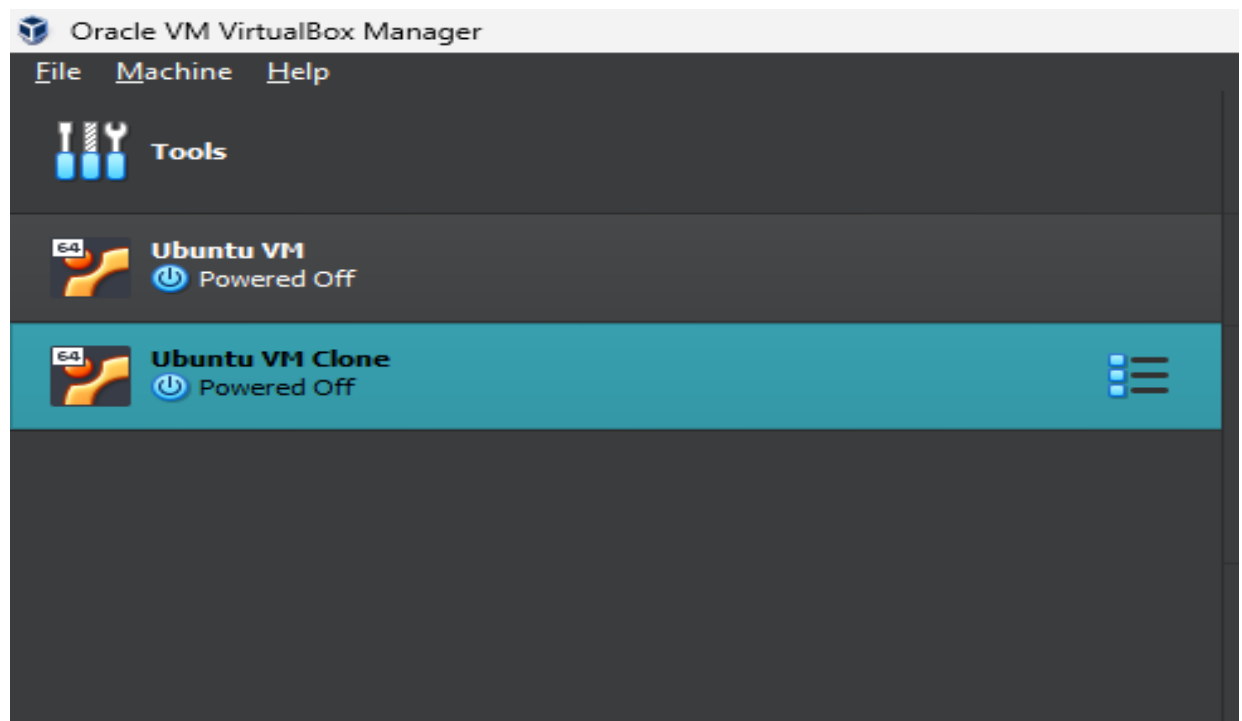
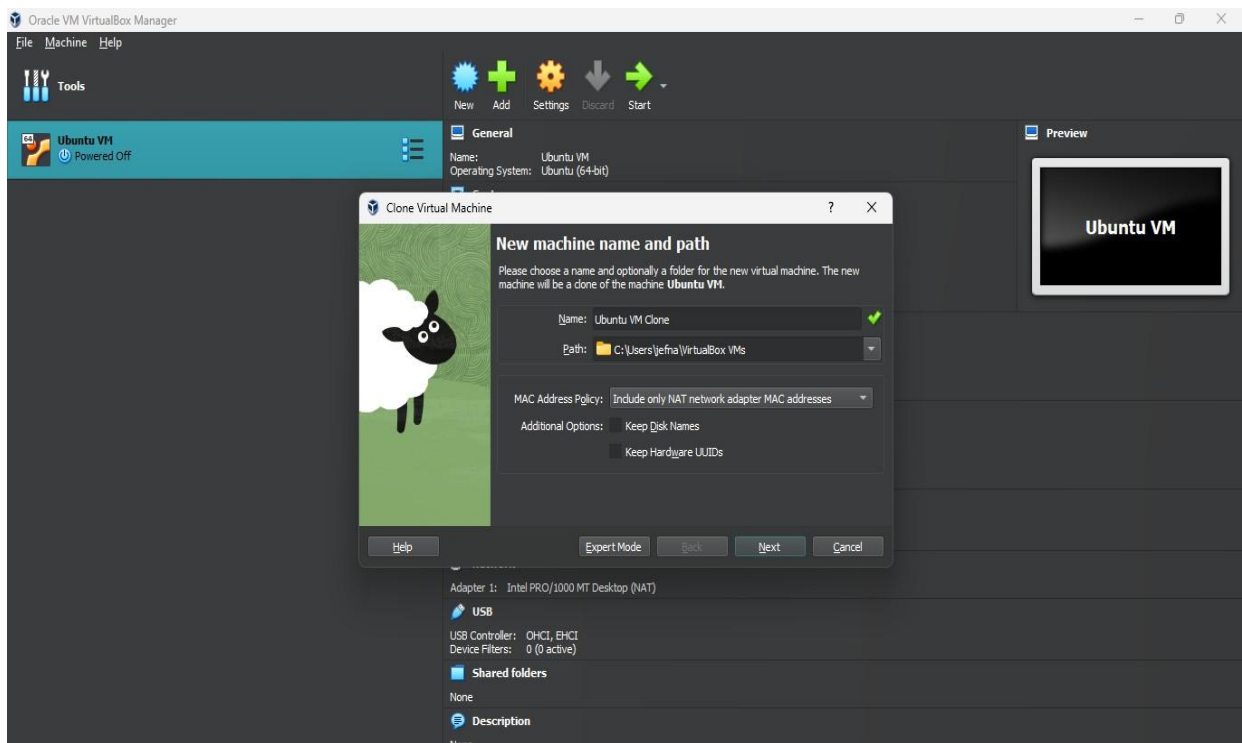
### **Aim**

**Clone your existing Ubuntu Virtual Machine and prove that the cloned VM works.**

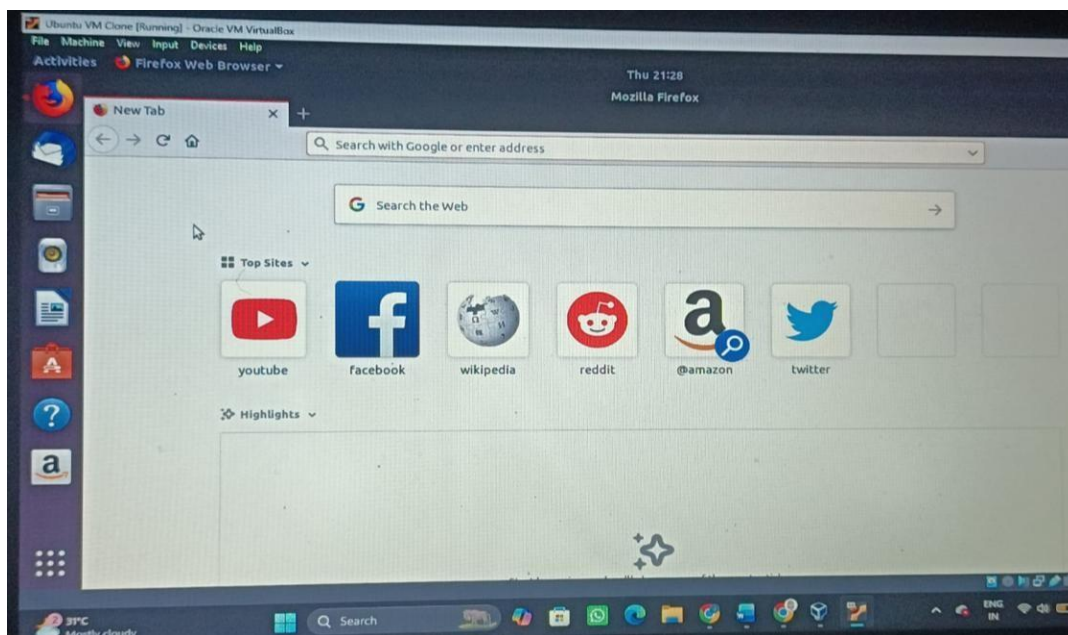
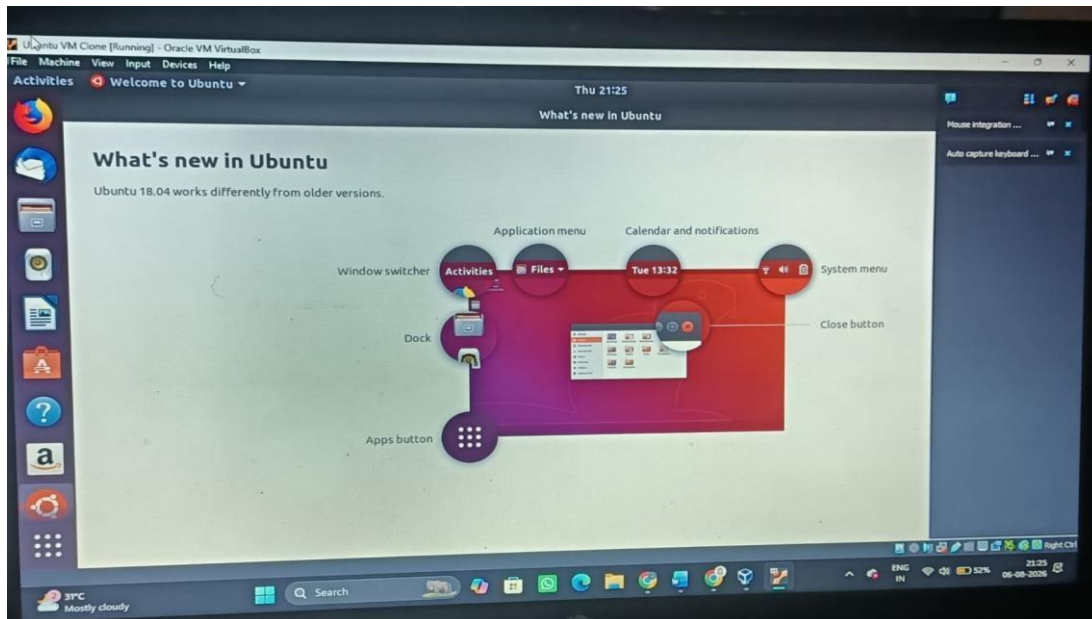
### **Procedure**

- 1. Open Oracle VM VirtualBox on the host operating system.**
- 2. Ensure the existing virtual machine (Ubuntu VM) is in the Powered Off state.**
- 3. Select the Ubuntu VM from the left panel.**
- 4. Click Machine → Clone (or right-click the VM and select Clone).**
- 5. Enter a name for the cloned virtual machine (e.g., Ubuntu Clone).**
- 6. Click Next and choose Full Clone as the cloning type.**
- 7. Keep the default MAC Address Policy and click Finish.**
- 8. Wait until the cloning process is completed successfully.**
- 9. Verify that the cloned virtual machine appears in the VirtualBox Manager.**
- 10. Select the Ubuntu Clone and click Start.**
- 11. Wait for the cloned VM to boot and verify that the operating system loads successfully.**
- 12. Shut down the cloned virtual machine after testing.**

Output:







## Result

The virtual machine was successfully cloned using Oracle VM VirtualBox, and the cloned VM was tested successfully by booting into the Ubuntu operating system.